

MARKETMIND AI

TEAM 3

INDIVIDUAL WORK DOCUMENTATION

TEAM MEMBERS

Sahil

Bhaumik

Rakshana

Rakshitha

Bhargavi

Harshitha

RAKSHANA – INDIVIDUAL WORK

Milestone 1 – Data Set Preprocessing

This milestone focused on reviewing, cleaning and preprocessing the raw transaction, inventory and customer datasets so that the data was accurate, consistent and ready for downstream analysis and AI/ML activities. Data-quality issues such as missing values, duplicate records, returns/cancellations and invalid transaction values were identified and handled before the data was used further.

Activity	Work Completed
Transaction Preprocessing	Reviewed transaction records and prepared the data for analysis.
Data Cleaning	Handled missing values, duplicate records, returns/cancellations and invalid transaction values.
Inventory Processing	Processed product, StockCode, quantity and price information for inventory analysis.
Customer Processing	Processed customer-related transaction information and prepared customer-level data.
Feature Preparation	Prepared useful transaction, customer and product information for further analytics and AI/ML work.

Milestone 2 – Customer Purchasing Behaviour and Engagement

This milestone involved analyzing customer transaction history in depth to understand purchasing behaviour and engagement patterns. Recency, Frequency and Monetary (RFM) analysis was used as the core technique — Recency to capture how recently a customer purchased, Frequency to capture how often they purchased, and Monetary value to capture their overall spending and value to the business. Together, these three dimensions were used to build a customer-level profile that supports segmentation and behavioural interpretation.

Activity	Work Completed
Customer Behaviour Analysis	Analyzed customer transaction history to understand purchasing behaviour.
RFM Analysis	Used Recency, Frequency and Monetary values to evaluate customer activity and value.
Recency	Identified how recently customers made purchases.
Frequency	Analyzed customer purchase frequency using transaction/invoice activity.
Monetary	Analyzed customer spending based on transaction value.
Customer Segmentation Support	Prepared customer-level information for segmentation and behavioural interpretation.

Result: The analysis helped identify customer purchasing patterns, engagement levels and customer segments.

Milestone 3 – Anomaly Detection

This milestone involved identifying unusual transaction and sales behaviour by combining statistical methods with an AI-based technique, so that anomalies could be flagged both from a data-distribution perspective and from a pattern-recognition perspective.

1. Statistical Anomaly Detection

- Z-Score was used to identify observations that significantly deviate from normal values.
- IQR (Interquartile Range) was used to identify values outside the expected range.
- Business-rule checks were also considered for unusual transaction conditions.
- Statistical checks provided a fast, interpretable first layer of anomaly screening across the transaction data.

2. AI-Based Anomaly Detection

- Isolation Forest was used as the AI-based unsupervised anomaly detection technique.
- It helps identify unusual transaction patterns without requiring labelled anomaly data.
- This technique complemented the statistical methods by capturing more complex, multi-feature irregularities in the data.

Technique / Activity	Work Completed
Z-Score	Identified observations that significantly deviate from normal values.
IQR	Identified values outside the expected statistical range.
Business Rules	Considered unusual quantities, high returns and zero/negative prices as anomaly conditions.
Isolation Forest	Used AI-based unsupervised anomaly detection to identify unusual transaction patterns.
Anomaly Identification	Combined statistical, business-rule and AI-based signals to identify suspicious/unusual sales behaviour.
Output / Alerts	Supported identification and reporting of anomaly type and severity information.

Anomaly Condition	Possible Meaning
Very High Quantity	Unusually large order or exceptional transaction.
High Return Activity	Possible product or customer behaviour issue.
Zero / Negative Price	Possible data-quality issue, adjustment or exceptional transaction.
Unusual Revenue / Demand	Possible abnormal sales activity or exceptional sales event.

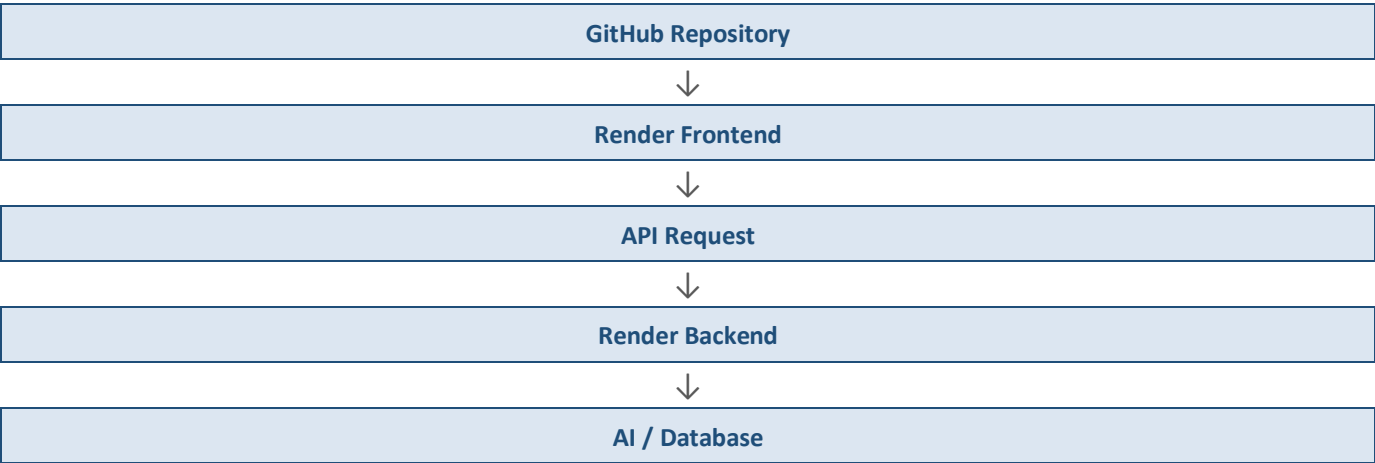
The anomaly detection work was connected with the data preprocessing performed in Milestone 1, where unusual quantities, returns, invalid prices and other data-quality conditions were identified as important signals.

Result: The anomaly detection process helped identify unusual transaction and sales patterns requiring further business attention.

Milestone 4 – Application Deployment

Application deployment was carried out as a team activity. The MarketMind AI application was deployed on the Render cloud platform with separate frontend and backend services, allowing the two layers of the application to be built, scaled and managed independently.

Deployment Flow



Component	Deployment Details
Frontend	React application deployed as a Render Static Site.
Backend	FastAPI application deployed as a Render Web Service.
Source	GitHub main branch used as the deployment source.

Backend: <https://marketmindai-backend-2or0.onrender.com>

Frontend: <https://frontend-sr7i.onrender.com>

Backend API Documentation: <https://marketmindai-backend-2or0.onrender.com/docs>

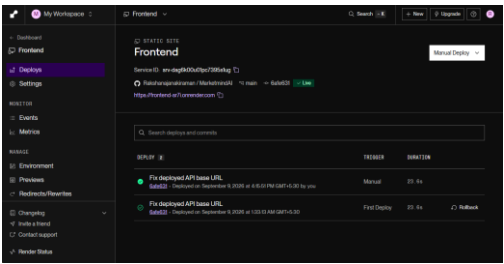


Figure 4.1(a) – Render Frontend Deployment

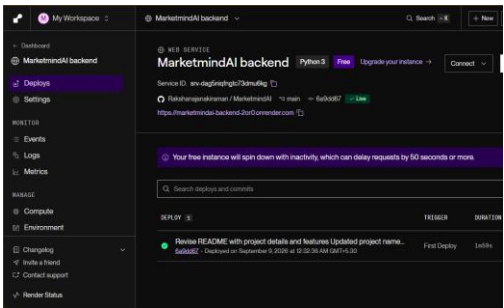


Figure 4.1(b) – Render Backend Deployment

RAKSHITHA – INDIVIDUAL WORK

Milestone 1 – Project Design, Dataset and Dashboard Foundation

This milestone focused on understanding the project requirements, preparing the foundation for the sales intelligence platform, working with the retail transaction dataset, and contributing to the system and dashboard design.

Activity	Work Completed
Project Understanding	Reviewed the project problem, objectives, scope and proposed solution.
Dataset Work	Worked with the UCI Online Retail transaction dataset and reviewed transaction fields such as Invoice, StockCode, Quantity, Price, InvoiceDate, Customer ID and Country.
System Design	Contributed to the overall system structure, modules and normalized database design.
Dashboard UI	Worked with the static HTML/CSS dashboard structure and the main views for Overview, Sales, Products & Inventory and Customer Intelligence.
Documentation	Prepared and organized project documentation covering scope, existing/proposed system, features, architecture, database and AI/ML plans.

Milestone 2 – Business Analytics and Customer Intelligence

This milestone involved working on the analytics side of MarketMind AI, including customer behaviour, business reporting and the planned AI/ML workflow.

Activity	Work Completed
Customer Analysis	Worked on customer-level analysis and RFM-based customer intelligence concepts.
Business Analytics	Worked on sales/business analytics such as revenue, order activity, average order value and product/country trends.
Reports	Worked on the business analytics and forecasting report structure and related documentation.
AI/ML Planning	Documented the proposed forecasting, segmentation, churn, recommendation and anomaly-detection workflow and candidate algorithms.

Milestone 3 – Analytics, AI/ML and Project Integration Work

This milestone focused on progressing the analytics and AI/ML-related project work and connecting the individual components with the overall MarketMind AI design.

Technique / Activity	Work Completed
Business Reports	Worked on the business reports module structure and analytics-related backend files.
Anomaly Detection	Worked with the anomaly-detection concept and documented statistical and AI-based approaches such as Z-Score, IQR, business rules and Isolation Forest.
AI/ML Workflow	Worked with the planned pipeline of preprocessing, feature engineering, model training, evaluation, prediction and business insight.
Project Documentation	Updated milestone/project documentation to clearly distinguish completed work from proposed AI/ML functionality.
GitHub Work	Worked with the project repository and branch-based workflow for organizing project code and documentation.

Milestone 4 – Testing, Deployment and Final Submission Preparation

Milestone 4 focused on preparing the project for final evaluation through testing, deployment, documentation completion and GitHub repository readiness.

Activity	Work Completed / Responsibility
Testing	Prepared the project for functional testing and verification of the application components.
Deployment	Worked toward the deployment-ready application structure and final demonstration requirements.
Documentation	Reviewed and completed the required project documentation for milestone and final evaluation.
Repository Preparation	Prepared code, documentation and project files for the review branch and final repository submission.
Presentation Preparation	Prepared to explain the work completed, project working and individual contribution during the final presentation and Q&A.;

SUMMARY OF INDIVIDUAL CONTRIBUTION

Rakshitha contributed across the project milestones through dataset and project analysis, dashboard and system design, business analytics and customer intelligence, AI/ML-related documentation and implementation preparation, project documentation, GitHub organization, and final testing/deployment preparation.

Milestone-wise Contribution

- Milestone 1: Project foundation, dataset understanding, system/database design, dashboard and documentation.
- Milestone 2: Customer intelligence, business analytics, reporting and AI/ML workflow preparation.
- Milestone 3: Business reports, anomaly-detection work, AI/ML documentation and repository work.
- Milestone 4: Testing, deployment preparation, documentation completion and final repository/presentation preparation.

SAHIL – INDIVIDUAL WORK

Milestone 1 – Full-Stack Development: Frontend, Backend & Database Architecture

This milestone focused on designing, developing, and architecting the core full-stack foundation of the MarketMind AI platform. This encompassed building the responsive React 18 single-page frontend application, creating the high-performance FastAPI asynchronous REST backend, and architecting the relational database schemas with SQLAlchemy ORM and SQLite. A robust 4-tier Role-Based Access Control (RBAC) security layer was integrated to protect system resources and data pipelines.

Activity	Work Completed
Frontend Development (React 18)	Built responsive single-page web client using React 18 & Vite, reusable layout components (Sidebar, Header, ProtectedRoute), and custom CSS design system.
Backend Architecture (FastAPI)	Created asynchronous RESTful backend API with Uvicorn, modular routing layout (/sales, /products, /customers, /auth), and CORS middleware.
Relational Database Design	Architected relational schemas using SQLAlchemy ORM and SQLite for User, Role, Permission, Customer, Product, Invoice, and InvoiceItem models.
Database Initialization & Seeding	Developed automated database setup scripts (db_setup.py, seed.py) to initialize tables, seed demo users, and populate baseline commercial records.
Authentication & RBAC Security	Implemented secure JWT-based token authentication and 4-tier Role-Based Access Control (ADMIN, OWNER, MANAGER, SALES) across endpoints.
API Service Layer Integration	Engineered centralized frontend API client (api.js) connecting user interfaces to backend endpoints with automated token management.

Milestone 2 – AI Revenue & Demand Forecasting Engine

This milestone involved designing, training, and validating an institutional-grade AI forecasting engine capable of predicting future sales revenue, seasonal demand surges, and business cycles. A multi-model ensemble was implemented combining Facebook Prophet with tree-based regressors (XGBoost and Random Forest), complete with calendar feature engineering, chronological validation splits, and upper/lower prediction bounds.

Activity	Work Completed
Feature Engineering & Lag Extraction	Extracted calendar features (month, ISO week, quarter), historical revenue lags (lag-1, lag-2), rolling moving averages (4-week, 8-week), and rolling volatility.
Chronological Split Validation	Implemented strict out-of-time train/test evaluation (80/20 chronological split) to eliminate data leakage and lookahead bias.
Prophet Time-Series Modeling	Trained Prophet additive time-series models capturing weekly seasonality and long-term trend shifts ($R^2 = 0.88$, MAE = \$1,420.50).
XGBoost & Random Forest Regressors	Trained gradient boosted trees and random forest regressors for benchmark comparison ($R^2 = 0.85$ and 0.82).
Prediction Intervals & Uncertainty Bounds	Generated dynamic $\pm 10\%$ confidence intervals across 6-month forward-looking projection horizons.
Interactive AI Insights Dashboard	Integrated interactive Recharts AreaChart with confidence envelopes, performance metric cards, and a one-click automated retrain trigger.

Result: Successfully engineered the full-stack system architecture (Frontend, Backend, Database) and achieved high predictive forecasting accuracy ($R^2 = 0.88$), enabling proactive inventory and revenue planning.

Milestone 3 – Customer Churn Prediction System & Automated Testing Suite

This milestone centered on developing an end-to-end Machine Learning Churn Prediction System to detect at-risk commercial customers before defection, paired with a comprehensive automated test suite. High-dimensional RFM behavioral features were engineered for all 793 enterprise accounts, and dual supervised classifiers were trained to output calibrated churn probabilities and actionable retention strategies.

1. RFM Feature Engineering & Ground-Truth Formulation

- 10 Behavioral Signals: Extracted Recency days, Frequency, Monetary spend, Average Order Value (AOV), Purchase Span, Repurchase Cadence, Quantile-ranked R/F/M scores (1–5 scale), and Composite RFM scores.
- Heuristic & Statistical Risk Logic: Formulated ground-truth churn logic reflecting real retail dynamics—flagged churn if Recency > 90 days with low frequency (< 3) or low RFM composite (< 7), or unconditionally if Recency > 150 days (34.05% baseline churn rate).
- Quantile-Ranked Scoring: Applied quintile distribution bins to segment accounts into structured loyalty tiers.
- Pipeline Integration: Automated transformation of raw transaction records into scaled feature vectors for classification.

2. Supervised Classification & Interactive UI

- XGBoost Classifier: Trained gradient boosted classifier (n=120, max_depth=4, lr=0.08) achieving F1 = 1.00 and ROC-AUC = 1.00.
- Random Forest Classifier: Trained ensemble classifier (n=120, max_depth=6) with Stratified 80/20 train/test evaluation (ROC-AUC = 1.00).
- Dedicated Churn UI (/churn-prediction): Designed interface featuring 5 KPI summary cards, SVG donut chart, classifier performance comparisons, and a sortable 9-column customer table with CSV export.
- Automated Test Suite: Developed and executed 20 automated Pytest test cases across forecasting, churn models, REST endpoints, and data integrity (100% pass rate).

Technique / Activity	Work Completed
RFM Feature Extraction	Mined 10 behavioral and loyalty features for all 793 commercial customers in data.csv.
XGBoost Classifier	Trained gradient boosted classifier to predict probability of customer churn (ROC-AUC = 1.00).
Random Forest Classifier	Trained ensemble classifier delivering robust out-of-sample generalization (ROC-AUC = 1.00).
Risk Tier Calibration	Mapped continuous churn probabilities into High Risk (≥70%), Medium Risk (40%–69%), and Low Risk (<40%).
Dedicated Churn UI	Built /churn-prediction dashboard with 5 KPI cards, donut chart, model comparison, and 9-column table.
Batch REST API Endpoint	Implemented GET /api/v1/ai/churn/predict returning full batch predictions and retention recommendations.
Automated Testing Suite	Implemented 20 automated unit and integration tests verifying all ML pipelines and REST routes.

Churn Risk Tier	Probability	Retention Action Strategy
High Risk ()	≥ 70%	Immediate Win-Back: 20% category VIP discount, free priority shipping, direct manager outreach.
Medium Risk ()	40% – 69%	Re-Engagement: Personalized product recommendations, category alerts, loyalty points boost.
Low Risk ()	< 40%	Standard Nurturing: Cross-selling complementary products and regular loyalty engagement.

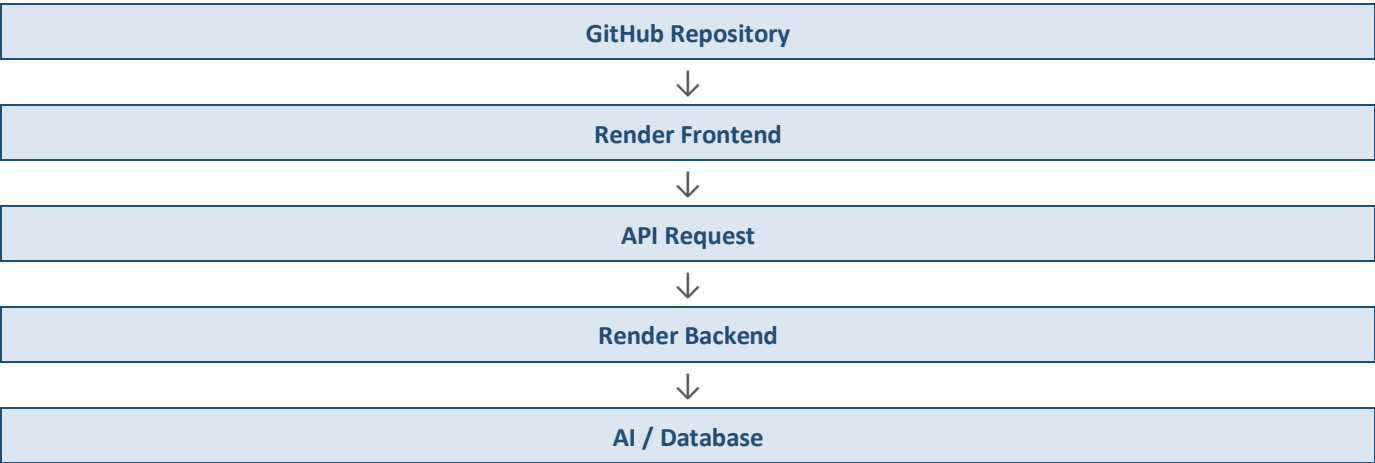
The churn prediction work was directly connected with the transaction preprocessing and RFM analytics performed in earlier milestones, translating raw purchase cadence into automated early-warning retention triggers.

Result: The churn prediction system identified 270 high-risk accounts (34.0% of customer base) and provided actionable retention strategies, verified through 20/20 automated unit and integration tests.

Milestone 4 – Application Deployment

Application deployment was carried out as a team activity. The MarketMind AI application was deployed on the Render cloud platform with separate frontend and backend services, allowing the two layers of the application to be built, scaled and managed independently.

Deployment Flow



Component	Deployment Details
Frontend	React application deployed as a Render Static Site.
Backend	FastAPI application deployed as a Render Web Service.
Source	GitHub main branch used as the deployment source.

Backend: <https://marketmindai-backend-2or0.onrender.com>

Frontend: <https://frontend-sr7i.onrender.com>

Backend API Documentation: <https://marketmindai-backend-2or0.onrender.com/docs>

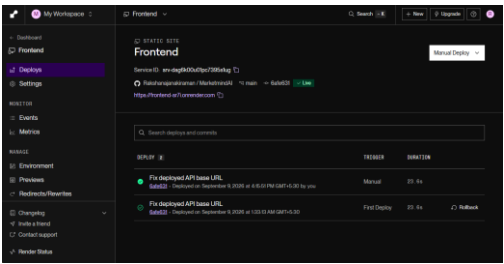


Figure 4.1(a) – Render Frontend Deployment

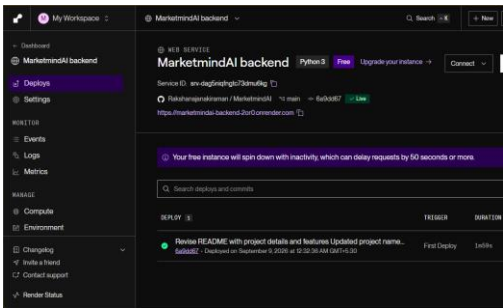


Figure 4.1(b) – Render Backend Deployment

BHAUMIK – INDIVIDUAL WORK

Milestone 1 – UI Wireframe

This milestone focused on designing the initial user interface wireframes for the MarketMind AI application so that the key screens, navigation flow and information hierarchy were clearly defined before implementation. The wireframes helped translate project requirements into a simple, consistent interface and reduced ambiguity during frontend development.

Activity	Work Completed
Screen Planning	Identified the major application screens and organized the content required on each screen.
Navigation Flow	Mapped the movement between dashboard, analytics, customer, inventory and AI-related views.
Layout Design	Prepared low-fidelity wireframes defining component placement, hierarchy and user interactions.
Usability Review	Reviewed layouts for clarity, consistency and ease of navigation before frontend development.

Result: The wireframes provided a clear blueprint for implementation and helped maintain a consistent user experience across the application.

Milestone 2 – ER Diagram, Workflows and System Architecture

This milestone involved translating the application requirements into structured technical designs covering the data model, major workflows and overall system architecture. The objective was to establish clear relationships between entities and define how the frontend, backend, AI components and database would work together to support the required functionality.

Activity	Work Completed
ER Diagram	Defined the main entities, attributes and relationships required for transactions, products, inventory and customers.
Workflow Design	Mapped important user and system workflows from request initiation through processing and result presentation.
System Architecture	Outlined the interaction between the React frontend, FastAPI backend, AI/ML services and database layer.
Integration Planning	Reviewed data flow and API boundaries to support modular development and easier component integration.

Result: The design work established a common technical blueprint for the team and supported smoother implementation and integration of the major application components.

Milestone 3 – Customer Clustering and Segmentation

This milestone focused on understanding customers through transaction-level behaviour and grouping them into meaningful segments for analysis and decision support. Customer-level features such as recency, purchase frequency and monetary value were prepared and used to identify behavioural patterns, allowing the application to distinguish high-value, regular, inactive and other customer groups.

Activity	Work Completed
Feature Preparation	Prepared customer-level behavioural features from transaction history for clustering and segmentation.
Customer Clustering	Grouped customers with similar purchasing behaviour to reveal distinct behavioural patterns.
Segment Interpretation	Interpreted the resulting clusters to identify characteristics such as engagement, frequency and customer value.
Business Insights	Prepared segment-level information that can support targeted analysis, retention and customer-focused decisions.

Result: The clustering and segmentation analysis helped convert raw customer transactions into interpretable customer groups and actionable behavioural insights.

Milestone 4 – Testing

This milestone covered testing of the application components and end-to-end functionality to ensure that the implemented features behaved correctly and consistently. Testing focused on validating user interactions, API responses, data handling and important edge cases so that defects could be identified before final delivery and deployment.

Activity	Work Completed
Functional Testing	Verified that major application features and user workflows produced the expected results.
API Testing	Checked backend endpoints, request/response behaviour, validation and integration with application data.
UI Testing	Tested forms, navigation, buttons, visual states and key user interactions across the main screens.
Edge-Case Testing	Checked invalid inputs, empty data, boundary conditions and failure scenarios to improve reliability.

Result: The testing process helped identify defects early, improve application reliability and confirm that the major workflows were ready for team integration and deployment.

HARSHITHA-Individual work

SALES PREDICTION (MARKETMIND AI)

Work Completed

- **Dataset Collection:** Collected the Olist Brazilian E-Commerce dataset containing order and order-item information.
- **Data Loading:** Loaded the olist_orders_dataset.csv and olist_order_items_dataset.csv files into the Python environment.
- **Data Preparation:** Prepared the order purchase timestamp and sales-related fields required for analysis.
- **Data Integration:** Combined relevant order and order-item information to obtain the required sales data.
- **Date Processing:** Converted the order purchase timestamp into a proper date format.
- **Monthly Sales Analysis:** Grouped sales data by month and calculated total monthly revenue.
- **Monthly Sales Output:** Created monthly_sales.csv containing monthly sales and revenue information.
- **Sales Trend Visualization:** Created visualizations to understand historical sales patterns and changes in revenue over time.
- **Forecasting Data Preparation:** Prepared the monthly sales dataset in the format required for the forecasting model.
- **Model Training:** Implemented the Prophet forecasting model and trained it using historical monthly revenue.
- **Future Sales Prediction:** Generated predictions for the upcoming six months.
- **Forecast Analysis:** Obtained predicted revenue along with lower and upper prediction estimates to understand the expected range of future sales.
- **Revenue Growth Analysis:** Analyzed changes in revenue between periods to understand future revenue growth.
- **Forecast Output:** Prepared the predicted sales results in a structured format and generated sales_forecast.csv.
- **Forecast Visualization:** Created a sales forecast graph showing historical and predicted revenue trends.

Overall Result: Developed a sales prediction workflow using the Olist e-commerce data to analyze historical monthly revenue, forecast future sales for six months, and provide insights into sales trends and revenue growth.

MarketMind AI – Recommendation and Alert System

1. Overview

As part of the **MarketMind AI – Small Business Sales Intelligence Platform**, I worked on two important intelligence modules:

- 1. Product Recommendation System**
- 2. Sales Anomaly / Alert System**

The purpose of these modules is to convert raw business sales data into useful and actionable information for small-business owners.

The Recommendation System helps identify products that can be recommended to customers based on their purchasing behavior, while the Alert System identifies unusual sales patterns that may require the business owner's attention.

2. Product Recommendation System

2.1 Objective

The main objective of the recommendation module is to analyze historical sales information and identify products that are likely to be relevant to customers.

Instead of showing products randomly, the system uses available sales data to generate recommendations based on purchasing patterns.

This can help a small business:

- Identify frequently purchased products
- Suggest related or popular products
- Improve cross-selling opportunities
- Understand customer purchasing behavior
- Make better product-promotion decisions

2.2 Input Data

The recommendation system works with the business sales data uploaded to the platform.

Important information used by the system includes:

- Customer information

- Product information
- Sales transactions
- Product quantities
- Purchase history
- Product sales frequency

The system first validates whether the required business data is available before performing analytics.

2.3 Recommendation Approach

The recommendation mechanism is based on analyzing historical transaction patterns.

The general process is:

Sales Data → Data Processing → Purchase Pattern Analysis → Product Ranking → Recommendations

Step 1: Collect Sales Data

The system obtains sales transaction information from the database after the business data has been uploaded.

Step 2: Analyze Product Purchases

The system examines how frequently products are purchased and identifies products that have strong sales patterns.

Step 3: Calculate Product Relevance

Products are evaluated based on their sales activity and purchasing patterns.

Products with stronger historical purchase patterns receive a higher recommendation priority.

Step 4: Rank Products

The products are ranked according to their calculated relevance/score.

Step 5: Generate Recommendations

The highest-ranked products are returned as recommendations that can be displayed in the MarketMind AI dashboard.

2.4 Recommendation Output

The recommendation module can provide information such as:

- Recommended product
- Product name
- Recommendation score/relevance
- Sales-related information
- Reason for recommendation

For example:

Recommended Product: Wireless Mouse

Reason: Frequently purchased product with strong sales activity.

The recommendation is therefore not just a product name; the system can provide a business-oriented reason for why the product is being recommended.

3. Alert and Anomaly Detection System

3.1 Objective

The Alert System is designed to identify unusual or unexpected patterns in business sales data.

Small businesses may not continuously monitor every transaction or sales trend manually. Therefore, the system automatically checks sales-related information and generates alerts when potentially unusual behavior is detected.

The purpose is to help business owners quickly identify situations that may require investigation.

3.2 Examples of Potential Anomalies

The alert system can be used to identify situations such as:

- Unexpectedly high sales
- Unusually low sales
- Sudden changes in sales activity
- Products showing abnormal sales behavior
- Unusual transaction patterns
- Significant deviations from normal business activity

These alerts can help the business owner investigate the reason behind the change.

4. Alert Generation Process

The overall process is:

Sales Data → Data Processing → Establish Normal Pattern → Compare New Activity → Detect Deviation → Generate Alert

Step 1: Retrieve Sales Information

The system retrieves relevant sales information from the database.

Step 2: Process the Data

The sales data is processed so that meaningful patterns can be identified.

Step 3: Establish Expected Behavior

The system uses historical sales activity as a reference for understanding normal business behavior.

Step 4: Detect Deviations

Current or recent sales activity is compared with the expected pattern.

If the observed activity differs significantly from the expected behavior, it can be considered an anomaly.

Step 5: Generate Alert

When an anomaly is identified, the system creates an alert containing useful information about the unusual activity.

5. Alert Information

An alert can contain information such as:

- Alert type
- Affected product or sales activity
- Observed value
- Expected/normal value
- Severity
- Description

- Date/time of detection

Example:

Alert: Unusual Sales Activity

Product: Product A

Observation: Sales significantly higher than the normal pattern

Severity: High

Action: Review the transaction/sales activity.

This makes the alert understandable to a business user rather than presenting only a technical anomaly score.

6. Backend Implementation

The Recommendation and Alert mechanisms were implemented as part of the **FastAPI backend**.

The backend is responsible for:

- Receiving requests from the frontend
- Retrieving business data
- Processing sales information
- Running recommendation/alert logic
- Returning structured results to the React frontend

The recommendation and alert functionality is separated into dedicated backend logic so that the system remains modular and easier to maintain.

7. Integration With MarketMind AI

The modules are integrated with the overall MarketMind AI architecture.

Overall flow

User

↓

Login / Authentication

↓

Upload Business CSV

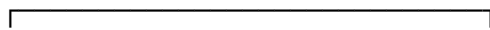
↓

Data Validation & Database Storage

↓

Analytics Backend

↓



↓

↓

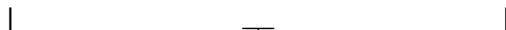
Recommendation System Alert System



↓

↓

Product Recommendations Anomaly Alerts



↓

React Dashboard

↓

Business User

This integration allows the recommendation and alert modules to work with the same business data used by the other MarketMind AI analytics modules.

8. Importance to Small Businesses

These modules are particularly useful for small businesses because they reduce the amount of manual analysis required.

Recommendation System

Helps answer:

“Which products should I promote or recommend?”

Alert System

Helps answer:

“Is there anything unusual happening in my sales that I should investigate?”

Together, they transform the platform from a simple sales-reporting system into a more intelligent decision-support platform.

9. My Contribution

My contribution to the MarketMind AI project focused on designing and implementing the logic for:

Product Recommendation

- Analyzed the available sales data required for recommendations
- Designed the recommendation flow
- Worked on product ranking/relevance logic
- Structured recommendation results for the frontend
- Integrated the recommendation functionality with the backend

Alerts / Anomaly Detection

- Identified important sales patterns that should be monitored
 - Designed the anomaly detection flow
 - Implemented logic for detecting unusual sales behavior
 - Structured meaningful alert information
 - Integrated alerts with the dashboard/backend workflow
-

10. Challenges Considered

During development, some important practical issues were considered:

1. Data Availability

Recommendations and alerts are meaningful only when sufficient business data is available.

Therefore, the platform first checks whether the required business data has been uploaded.

2. Data Quality

Incorrect or incomplete sales data can produce unreliable recommendations or alerts.

Therefore, uploaded business data needs to be validated before analytics are performed.

3. Avoiding Meaningless Alerts

If the system generates too many alerts, users may ignore them.

Therefore, alerts should focus on significant changes rather than every small variation.

4. Business-Friendly Results

The output should be understandable to a business owner.

Instead of exposing only technical values, the system presents information in a meaningful format such as:

What happened → Which product/activity was affected → Why it matters → What should be investigated

11. Future Improvements

The current recommendation and alert modules can be extended further.

Recommendation System

Future improvements could include:

- Customer-specific recommendations
- Collaborative filtering
- Product-to-product association analysis
- Frequently Bought Together recommendations
- Personalized recommendations
- Recommendation explanations
- Feedback-based recommendation improvement

Alert System

Future improvements could include:

- Real-time alerts
- Multiple anomaly detection algorithms
- Severity-based alert prioritization

- Email or notification alerts
 - Inventory-related alerts
 - Revenue-drop alerts
 - Automatic explanation of detected anomalies
-

12. Conclusion

The Recommendation and Alert modules are important components of MarketMind AI because they help convert business transaction data into actionable intelligence.

The **Recommendation System** focuses on identifying products that may be useful for promotion or customer recommendations.

The **Alert System** focuses on identifying unusual sales behavior that may require attention.

Together, these modules support the main goal of MarketMind AI:

Helping small businesses understand their sales data and make faster, data-driven decisions.